

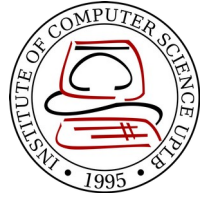
CMSC 137

Data Communications and Networking

Lecture Module
25 – Other Wireless Networks

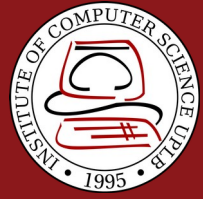


University of the Philippines
LOS BAÑOS



Module 25: Other Wireless Networks

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Lecturer



Cellular Networks: Overview

Overview



- Designed to provide communications between two moving units, called **mobile stations** or between one mobile unit and one stationary unit, called **land unit**
- A **service provider** must be able to locate and track caller, assign a channel to the call, and transfer the channel from base station to base station as the caller moves out of range

Overview



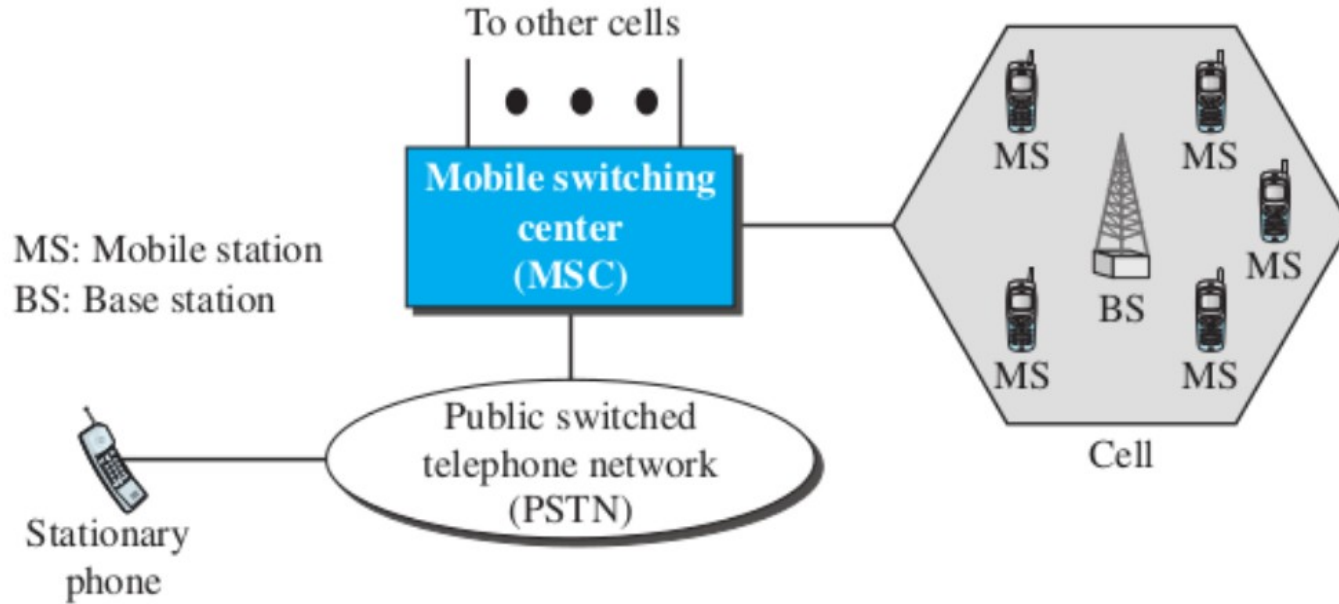
- To enable tracking, cellular service area is divided into small regions called **cells**
- Each cell has an antenna powered by a network station, called the **base station**
- Each base station is controlled by a switching office, called a **Mobile Switching Center (MSC)** to coordinate communication between all base stations and the telephone central office

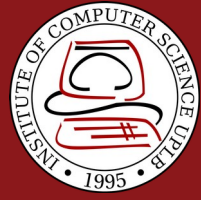
Overview



- Cell size is not fixed, typical radius of 1 to 12 miles
- Cell size is optimized to prevent interference of adjacent cell signals
- Transmission power of each cell is kept low to prevent interfering with the signal

Overview



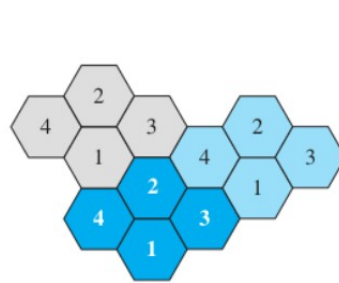


Cellular Networks: Operation

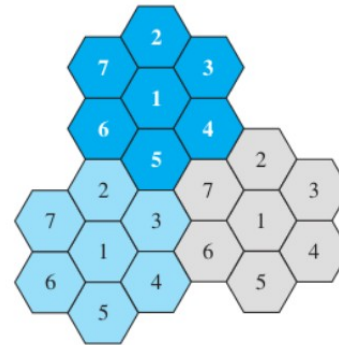
Frequency Reuse Principle



- Number of cells define the pattern, cells with same number in a pattern can use the same set of frequencies



a. Reuse factor of 4



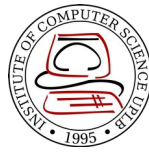
b. Reuse factor of 7

Transmitting



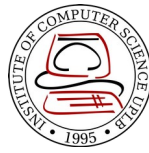
- Caller enters a code of 7 or 10 digits (**mobile number**)
- Mobile stations scans the band seeking a **setup channel** with a strong signal and sends the phone number to the closest base station using the channel
- Base station relays the mobile number to the MSC, MSC sends data to the telephone central office
- If called party is available, MSC assigns an unused voice channel to the call and the connection is established

Receiving



- Telephone central office sends the number to the MSC
- MSC searches for the location of the mobile station by sending query signals to each cell, called **paging**
- Once the mobile station is found, MSC transmits a ringing signal
- When mobile station answers, voice channel is assigned to the call allowing voice communication to begin

Handoff

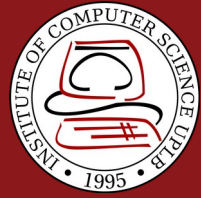


- During conversation, mobile station may move from one cell to another – signal may become weak
- MSC monitors signal level every few seconds, if signal is weak MSC finds a new cell then transfers the call to the new channel
- **Hard handoff** – mobile station only communicates with one base station
- **Soft handoff** – mobile station can communicate to both base stations in the old and new cells

Roaming



- Service provider has limited coverage
- Neighboring service providers can provide extended coverage through a **roaming contract**



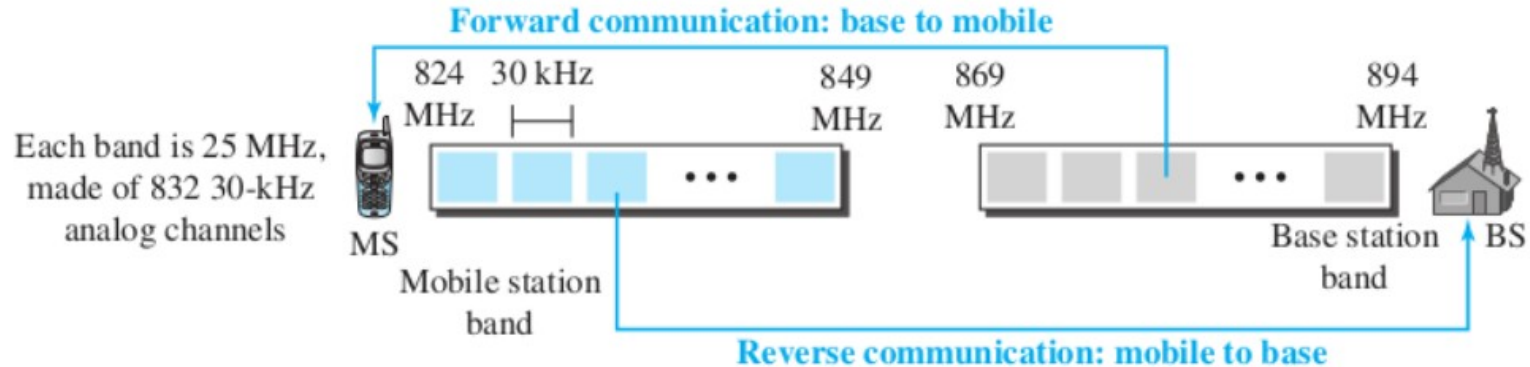
Cellular Networks: First Generation (1G)

- **Advanced Mobile Phone System (AMPS)** – used in North America, uses FDMA
- Designed for voice communication using analog signals
- Uses ISM 800-MHz band
 - **Forward channel** (base to mobile) - (869-894 MHz)
 - **Reverse channel** (mobile to base) - (824-849 MHz)

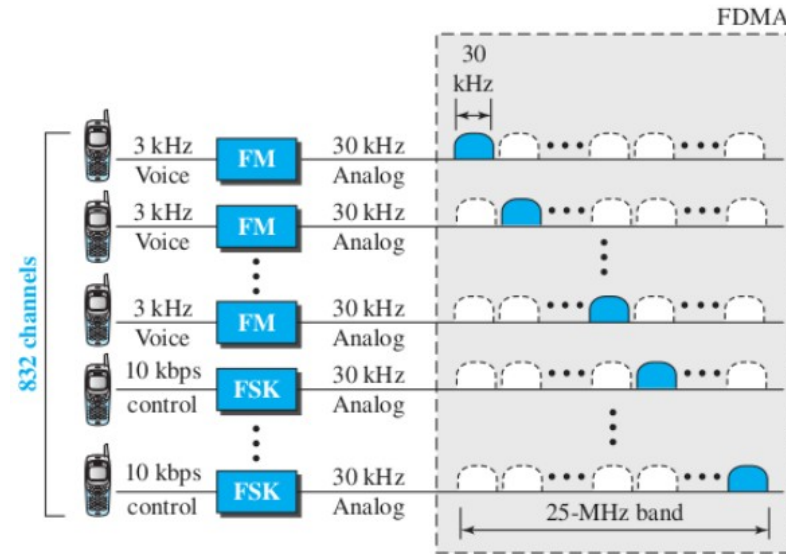
AMPS

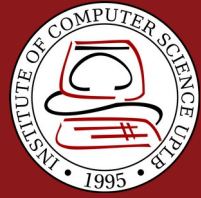


- 21 channels are used for control
- Frequency reuse factor is 7



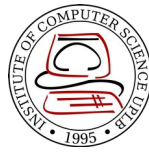
- Uses FM(voice channel) and FSK(control channel)





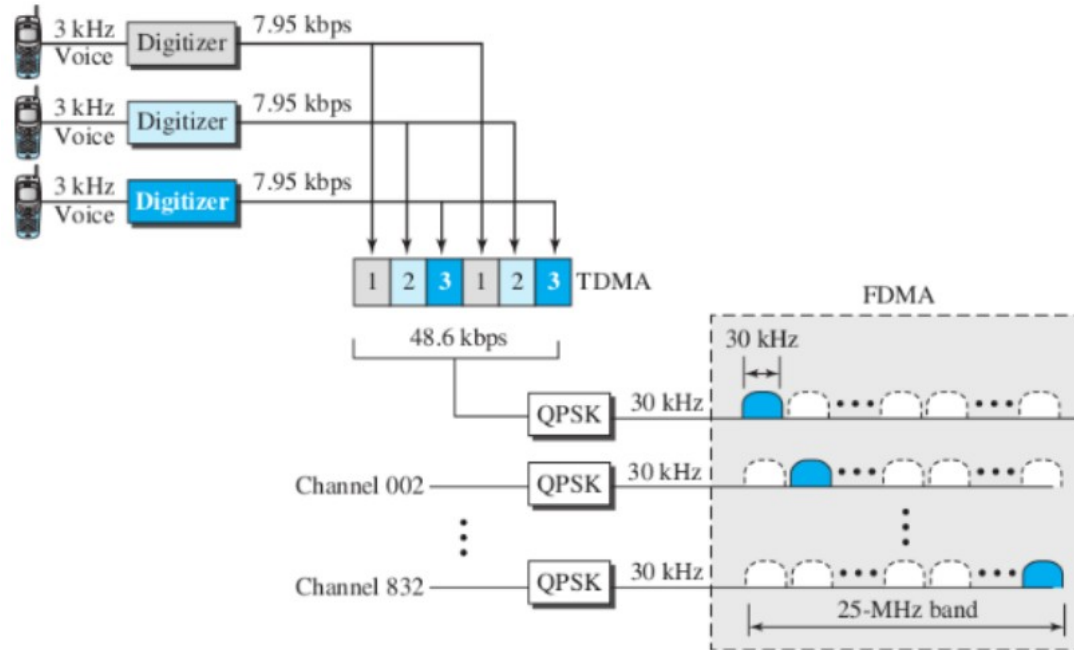
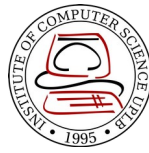
Cellular Networks: Second Generation (2G)

2G

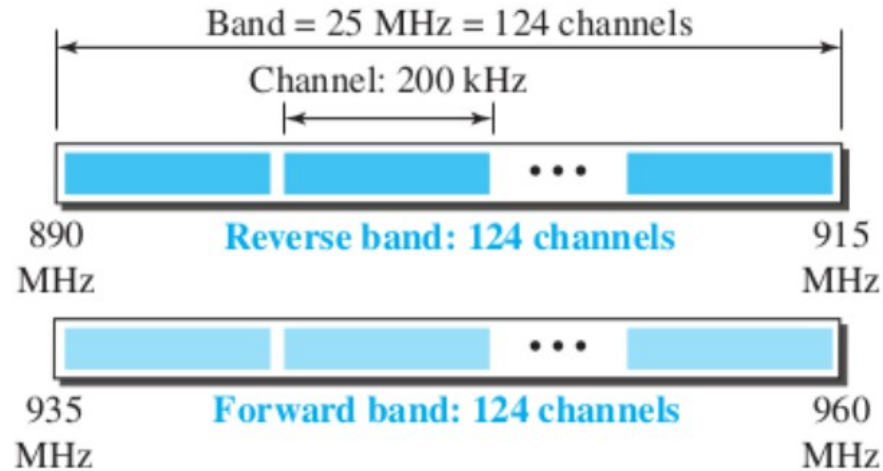


- Designed for digitized voice communications
- Systems: D-AMPS, GSM, IS-95

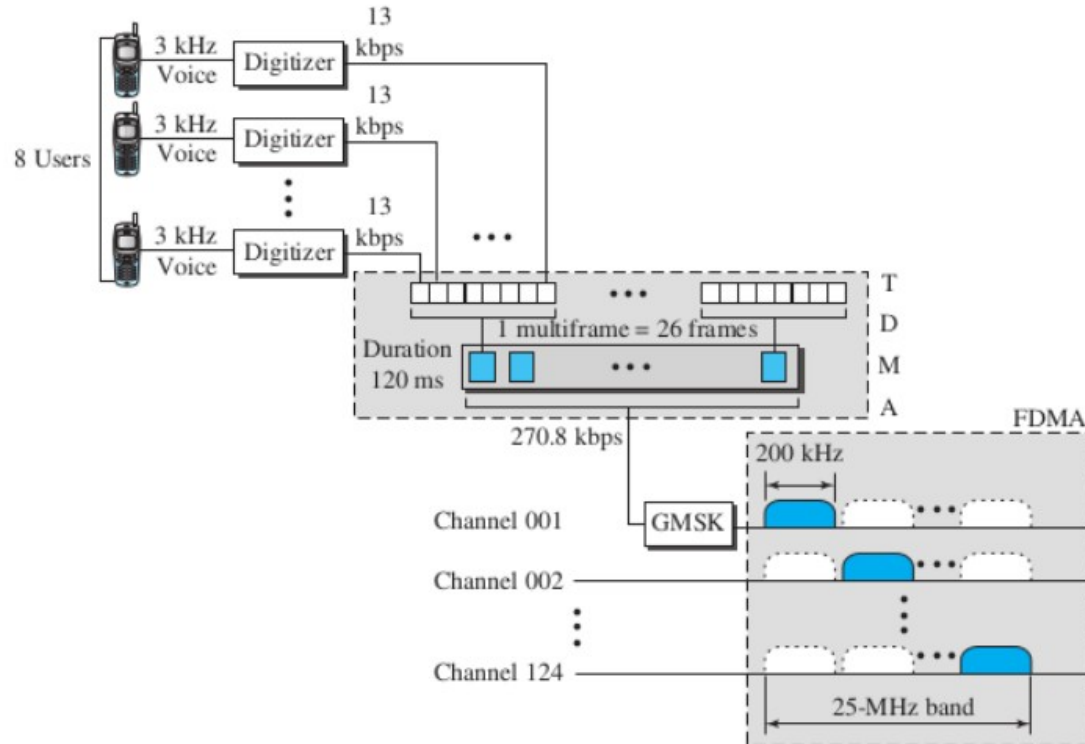
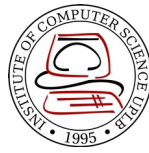
D-AMPS



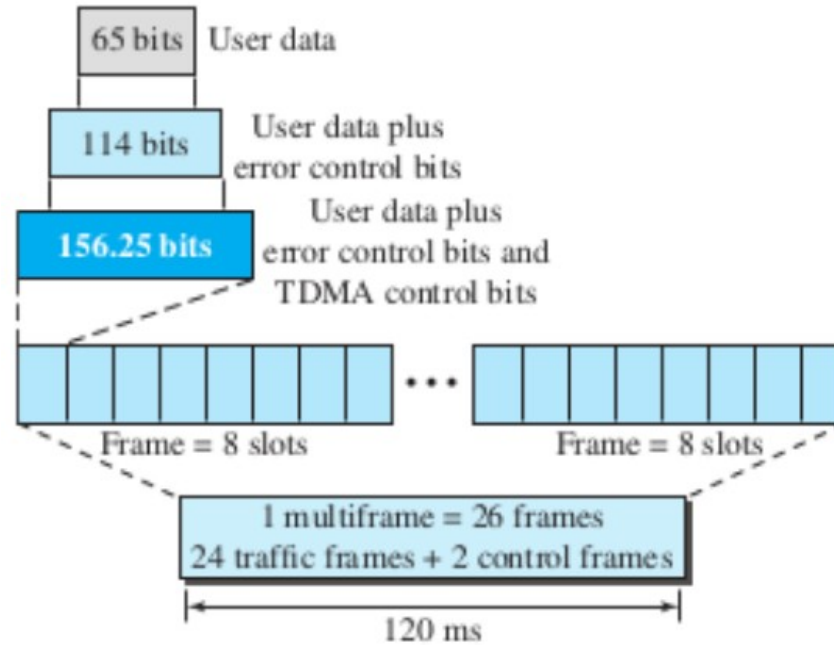
- **G**lobal **S**ystem for **M**obile Communication (GSM) – Europe-based
- Reuse factor: 3

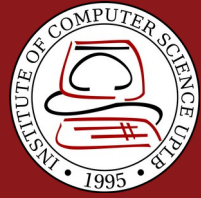


GSM



- Overhead



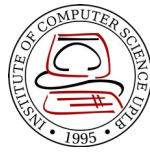


Cellular Networks: Third Generation (3G)



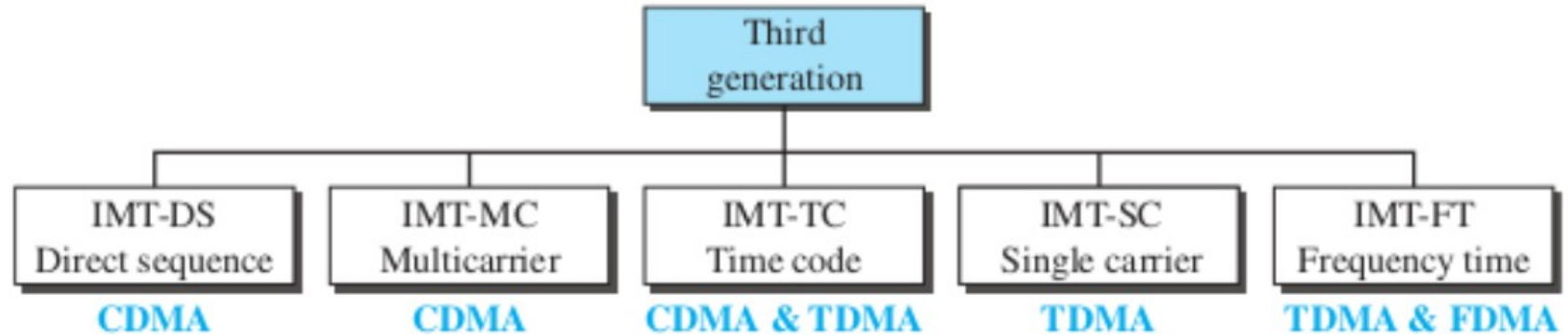
- Provide both digital data and voice communication
- Started in 1992, Internet Mobile Communication 2000 (IMT-2000)

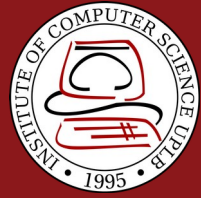
IMT-2000 Blueprint



- Voice quality comparable to that of the existing PSTN
- Data rate of 144 kbps for moving car, 384 kbps for pedestrians, 2 mbps for office
- Support for **packet-switched** and **circuit-switched** data services
- A band of 2 GHz
- Bandwidths of 2 MHz
- Interface to the Internet

IMT-2000 Radio Interfaces





Cellular Networks: Fourth Generation (4G)

4G Objectives

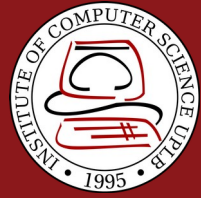


- Spectrally efficient system
- High network capacity
- 100Mbps for moving car, 1Gbps for stationary users
- Smooth handoff across heterogeneous networks
- Seamless connectivity and global roaming
- All IP, packet-switched networks

4G Features

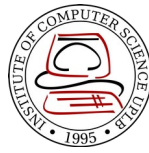


- Access Scheme
 - Orthogonal FDMA (UMTS), Interleaved FDMA (UMTS), Multicarrier CDMA (IEEE 802.20)
- Modulation
 - 64-QAM for LTE
- Radio System
 - Software Defined Radio (SDR)
- Antenna
 - Multiple-input Multiple-output (MIMO), Mutiuser MIMO



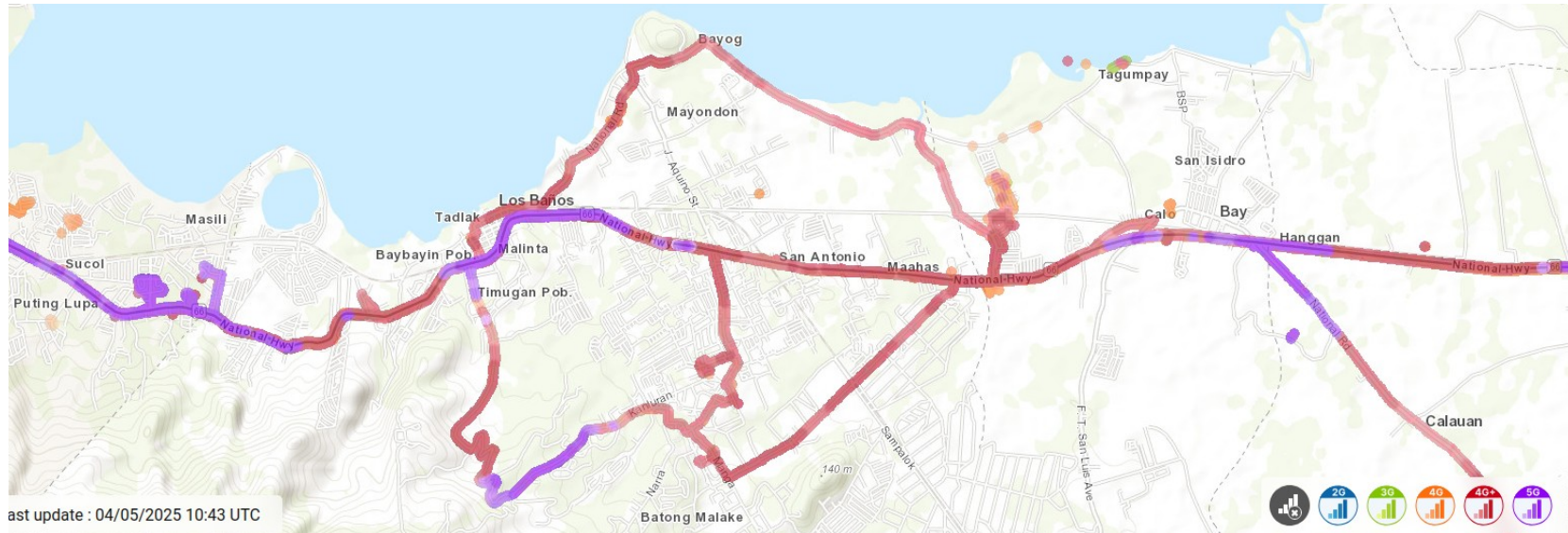
Cellular Networks: Fifth Generation (5G)

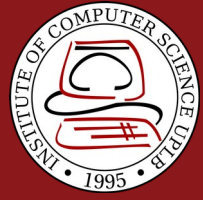
5G Features



- Uses Orthogonal FDM encoding
- Uses millimeter waves (wavelength)
- limited range, can be blocked by materials like walls and windows
- Health concerns?
- Can use low, medium, or high frequency bands (high frequency bands, 24-47 GHz, can achieve peak at 20 Gbps data rate)
- Internet of Things
- Useful for urban areas

<https://www.nperf.com/en/map/PH/-/1999180.SmartSunTNT-Mobile/signal?ll=14.180933667524748&lg=121.24717712402345&zom=14>





Satellite Networks

Overview

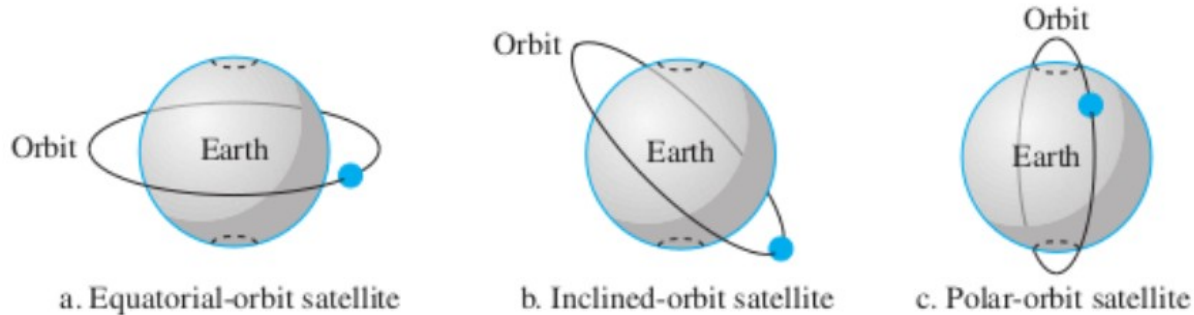


- Combination of nodes: **artificial satellites, Earth stations, end-user terminals or phones**
- Similar to cellular networks, Earth is divided into cells
- Useful in underdeveloped regions

Orbits



- Period – time required for a satellite to make a complete trip around Earth
 - $\text{Period} = \frac{1}{100} \times \text{distance} \times 1.5$
 - Earth's radius is 6378 Km



Footprint



- Process microwaves with bidirectional antennas (line-of-sight)
- **Footprint** is the specific area where the signal is aimed and the signal strength is maximum

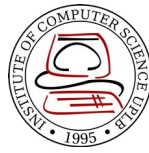


Frequency Bands



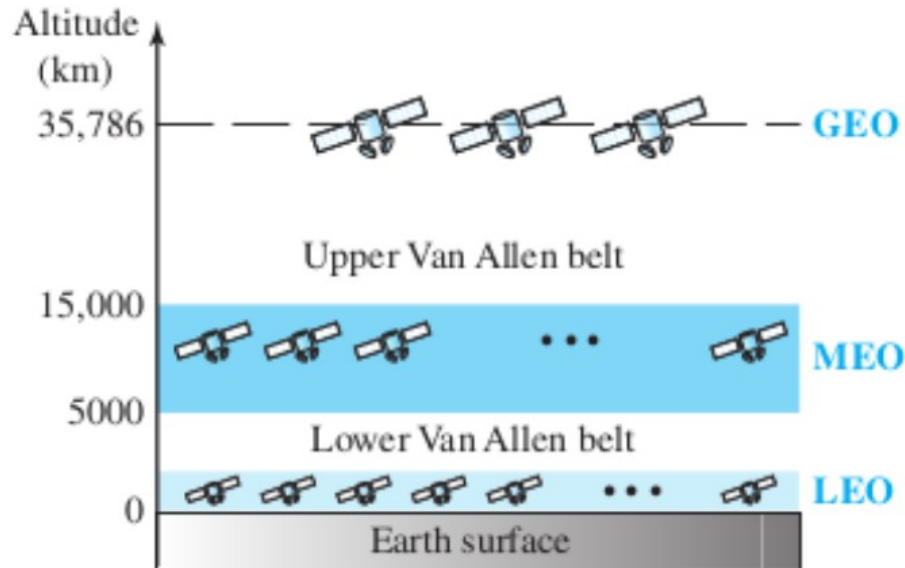
<i>Band</i>	<i>Downlink, GHz</i>	<i>Uplink, GHz</i>	<i>Bandwidth, MHz</i>
L	1.5	1.6	15
S	1.9	2.2	70
C	4.0	6.0	500
Ku	11.0	14.0	500
Ka	20.0	30.0	3500

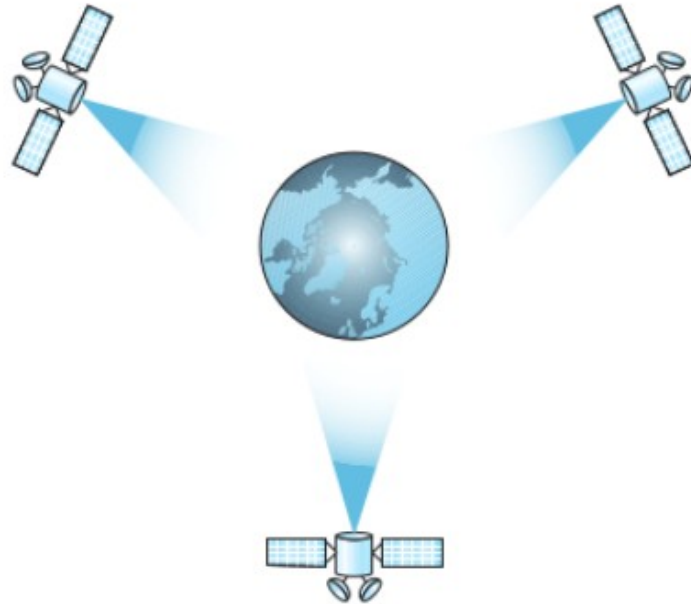
Satellite Categories



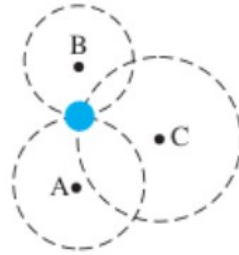
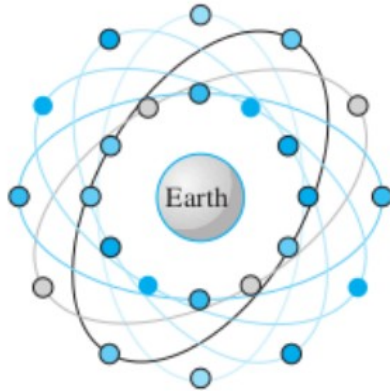
- **Geostationary Earth Orbit (GEO)** – 35786 Km
 - Typical satellite networks
- **Medium Earth Orbit (MEO)** – 5000-15000 Km
 - Example GPS
- **Low Earth Orbit (LEO)** – below 2000 Km

Satellite Categories

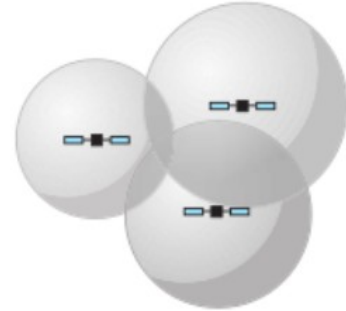




- Global Positioning System

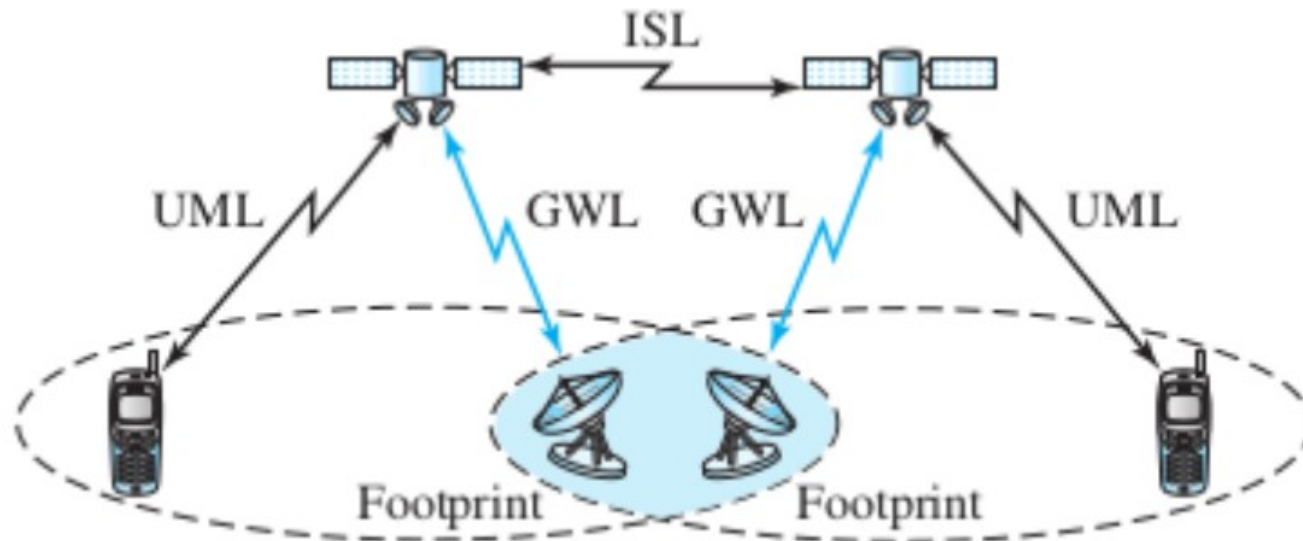


a. Two-dimensional trilateration

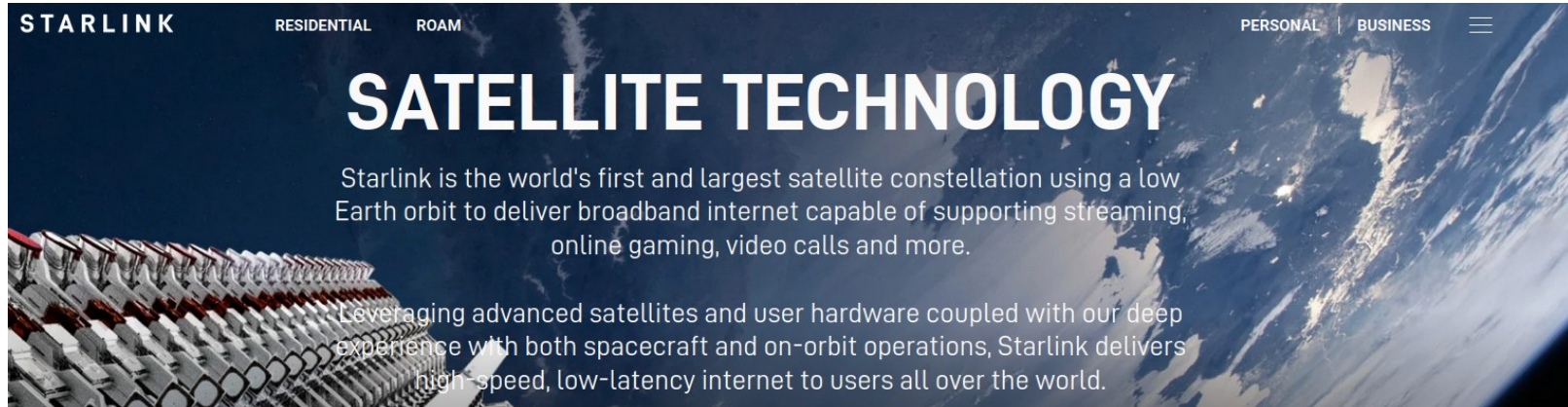


b. Three-dimensional trilateration

- Altitude between 500 and 2000 Km
- Satellite Speed: 20000 to 25000 Km/h
- Footprint diameter: 8000 Km
- Uses a constellation of Satellites connected with each other through **intersatellite links (ISL)**
- Mobile system connects through a **user mobile link (UML)**
- Satellite communicate with Earth station using a **gateway link (GWL)**



<https://www.starlink.com/technology>

A screenshot of the Starlink website's header. The background is a composite image showing a close-up of a satellite's solar panel array on the left and a view of Earth from space on the right. The text is overlaid on this background.

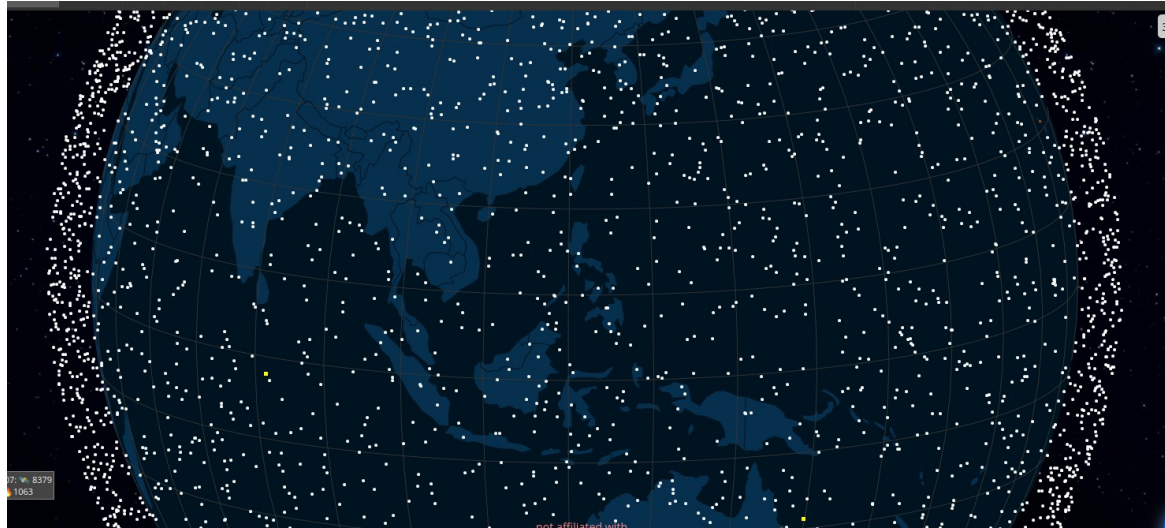
STARLINK RESIDENTIAL ROAM PERSONAL BUSINESS

SATELLITE TECHNOLOGY

Starlink is the world's first and largest satellite constellation using a low Earth orbit to deliver broadband internet capable of supporting streaming, online gaming, video calls and more.

Leveraging advanced satellites and user hardware coupled with our deep experience with both spacecraft and on-orbit operations, Starlink delivers high-speed, low-latency internet to users all over the world.

<https://satellitemap.space/?constellation=starlink>



References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.